

Mathematics A

Unit 2 • Chapter OL-T06 Classified

Cross-session • chapter-organised candidate

29 classified items • 102 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T06 • 29 items • 102 marks

Chapter	Items	Marks	Starts
01. OL-T06 Percentages	29	102	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Percentages

Unit 2 • 102 marks • 29 item(s)

June 2025 | 4MA1-1H Q2 | 4 marks

Profit, loss or percentage margin from costs and revenue

June 2025 | 4MA1-1H Q5 | 3 marks

Reverse a decrease or sale discount

June 2025 | 4MA1-1H Q13 | 3 marks

Compare or invert repeated percentage change

June 2025 | 4MA1-2H Q7 | 3 marks

Repeated growth or decay with known rate and time

June 2024 | 4MA1-1H Q4 | 3 marks

Reverse an increase or changed population/value

November 2023 | 4MA1-1H Q3 | 4 marks

Profit, loss or percentage margin from costs and revenue

November 2023 | 4MA1-1H Q6 | 4 marks

Reverse a decrease or sale discount

June 2023 | 4MA1-1H Q3 | 4 marks

Profit, loss or percentage margin from costs and revenue

November 2024 | 4MA1-1H Q7 | 4 marks

Reverse a decrease or sale discount

November 2024 | 4MA1-2H Q4 | 2 marks

Reverse an increase or changed population/value

November 2024 | 4MA1-2H Q4 | 3 marks

Calculate a percentage increase or decrease

June 2022 | 4MA1-1H Q6 | 3 marks

Reverse an increase or changed population/value

June 2022 | 4MA1-1H Q13 | 5 marks

Find a percentage of an amount

June 2022 | 4MA1-2H Q8 | 5 marks

Combine percentage with fraction or ratio groups

January 2022 | 4MA1-1H Q4 | 5 marks

Profit, loss or percentage margin from costs and revenue

... 14 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

- 2 Anna makes cups.
Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.
Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

- (a) Work out the percentage profit Anna makes.
Show your working clearly.

..... %
(4)

WORKING SPACE

- 5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.
Work out the normal price of the watch.

..... euros

(Total for Question 5 is 3 marks)

WORKING SPACE

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%

In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by $x\%$

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

WORKING SPACE

- 7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

(Total for Question 7 is 3 marks)

WORKING SPACE

Question 03

4MA1-1H Q4

MODULAR UNIT 2

3 marks

- 4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.

This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

..... kilometres

WORKING SPACE

Question 01

4MA1-1H Q3 demand

MODULAR UNIT 2

4 marks

3 Joshua buys a car for \$12 500

He sells the car to Nina.

Nina pays

- a deposit of \$1500
- followed by 36 monthly payments of \$450

Work out Joshua's percentage profit.

.....%

WORKING SPACE

Question 02

4MA1-1H Q6 demand

MODULAR UNIT 2

4 marks

- 6 Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket **A**
 $\frac{1}{6}$ off normal price

ticket **B**
20% off normal price

The sale price of ticket **A** is 140 euros.

The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

WORKING SPACE

Question 03

4MA1-1H Q3

MODULAR UNIT 2

4 marks

- 3 Nanette buys 60 notebooks for a total cost of 780 dirhams.

Nanette sells 70% of the notebooks for 22 dirhams each.

She sells the remaining notebooks for 19 dirhams each.

Work out Nanette's percentage profit.

Give your answer correct to 3 significant figures.

..... %

WORKING SPACE

Question 06

4MA1-1H Q7 M01

MODULAR UNIT 2

4 marks

- 7 Shop **A** and Shop **B** have offers for buying the same type of laptop.

The normal price of the laptop in Shop **A** is different to the normal price of the laptop in Shop **B**

Shop A	Shop B
Our normal price £475	Get 15% off our normal price
Get 16% off our normal price	Only pay £408

Which shop gives more money off their normal price?
Show your working clearly.

WORKING SPACE

Question 18

4MA1-2H Q4 M01

MODULAR UNIT 2

2 marks

- 4 N is a number.
17% of N is 357

(a) Work out the value of N

$N = \dots\dots\dots$
(2)

WORKING SPACE

Question 19

4MA1-2H Q4 M02

MODULAR UNIT 2

3 marks

In 2019, the population of a village was 650

In 2020, the population of the village was 806

(b) Work out the percentage increase in the population.

..... %
(3)

WORKING SPACE

Question 06

4MA1-1H Q6 Q6(a)

MODULAR UNIT 2

3 marks

- 6 Asha bought an apartment.

The table gives information about the value of apartments, in euros, and the annual service charge band.

Value (x euros)	Service charge band
$x \geq 700\,000$	A
$600\,000 \leq x < 700\,000$	B
$500\,000 \leq x < 600\,000$	C
$400\,000 \leq x < 500\,000$	D
$0 < x < 400\,000$	E

In 2021, the value of Asha's apartment was 634 400 euros.

The value of Asha's apartment had increased by 4% from its value in 2020

- (a) Has the annual service charge band changed for Asha's apartment?
Show your working clearly.

(3)

Working space for Question 06

Continue your method

UNIT 2**3 marks**

WORKING SPACE

Question 12

4MA1-1H Q13 Q13

MODULAR UNIT 2

5 marks

13 Antoine is going on holiday.

He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Antoine pays to the holiday company.

Payment	Amount paid
Payment 1	$\frac{3}{8}$ of the total cost
Payment 2	45% of the total cost
Payment 3	\$406

Work out how much Antoine has to pay for Payment 2

\$

Working space for Question 12

Continue your method

UNIT 2

5 marks

WORKING SPACE

Question 22

4MA1-2H Q8 Q8

MODULAR UNIT 2

5 marks

- 8 Behnaz makes 300 celebration cards so that

number of birthday cards : number of anniversary cards : number of congratulations cards = 7:5:3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.

Give your answer in its simplest form.

WORKING SPACE

Question 03

4MA1-1H Q4

MODULAR UNIT 2

5 marks

- 4 Sarah makes and sells mugs.

One day she makes 150 mugs.

Her total cost for making these mugs is £1140

Of these mugs

$\frac{2}{5}$ are small mugs

32% are medium mugs

and the rest are large mugs

Here is Sarah's price list for selling each mug.

MUGS	
Small	£8.50
Medium	£11.20
Large	£14.20

Sarah sells all 150 mugs.

Work out her percentage profit.

Give your answer correct to the nearest whole number.

.....%

Working space for Question 03

Continue your method

UNIT 2**5 marks**

WORKING SPACE

Question 21

4MA1-2H Q8 Q8(a)

MODULAR UNIT 2

2 marks

- 8 The table gives information about the average house price in England in 2018 and in 2019

Year	2017	2018	2019
Average house price (£)		228 314	231 776

- (a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

..... %
(2)

WORKING SPACE

Question 22

4MA1-2H Q8 Q8(b)

MODULAR UNIT 2

3 marks

The average house price in 2019 was 7.7% greater than the average house price in 2017

- (b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£
(3)

WORKING SPACE

- 8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says,

“The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017”

- (a) Is Brigitte correct?

You must show how you get your answer.

(4)

YOUR WORKING

Henri wants to insure his car.

He gets a discount of 15% off the normal price.

Henri pays 952 euros for his car insurance after the discount.

(b) Work out the discount that Henri gets.

..... euros
(3)

YOUR WORKING

3 Gladys buys a table for \$465 to sell in her shop.

She sells the table for \$520

- (a) Work out the percentage profit that Gladys makes from the sale of the table.
Give your answer correct to 3 significant figures.

..... %
(3)

YOUR WORKING

Gladys has a sale in her shop.

She decreases all the normal prices by 12%

The normal price of an armchair was \$550

(b) Work out the sale price of the armchair.

\$.....
(3)

YOUR WORKING

6 Mariana sells bags of bird food.

The bags that Mariana sold last week each contained 12 kg of seeds.

The bags that she is going to sell next week will each contain a mixture of nuts and seeds where for each bag

$$\text{weight of nuts : weight of seeds} = 4 : 5$$

The total weight of the nuts and the seeds in each bag will be 19.35 kg

The weight of seeds in each bag that Mariana sells next week will be less than the weight of seeds in each bag that Mariana sold last week.

Work out this decrease as a percentage of the weight of seeds in each bag that Mariana sold last week.

Give your answer correct to one decimal place.

..... %

YOUR WORKING

11 Chengbo sold a house for 180 000 yuan.

The amount for which he sold the house is 24% more than the amount he paid for the house.

- (a) Work out how much Chengbo paid for the house.
Give your answer correct to 3 significant figures.

..... yuan
(3)

YOUR WORKING

Zhi bought a house on 1st January 2017
When she bought the house, its value was 120 000 yuan.

The value of the house increased by 1.8% per year.

- (b) Work out the value of Zhi's house on 1st January 2020
Give your answer correct to 3 significant figures.

..... yuan
(3)

YOUR WORKING

- 8 In 2018, the population of Sydney was 5.48 million.
This was 22% of the total population of Australia.

Work out the total population of Australia in 2018
Give your answer correct to 3 significant figures.

..... million

YOUR WORKING

- 6 Victor buys 12 bottles of apple juice for a total cost of \$21
Victor sells all 12 bottles at \$2.45 each bottle.

Work out Victor's percentage profit.

..... %

YOUR WORKING

9 In a sale, normal prices are reduced by 20%

A bag costs 1080 rupees in the sale.

Work out the normal price of the bag.

..... rupees

(Total for Question 9 is 3 marks)

YOUR WORKING

6 Shane bought a car.

The amount Shane paid for the car was \$32 000

Theresa also bought a car.

To pay for this car, Theresa paid a deposit of \$18 000 together with 14 monthly payments of \$1160

Theresa paid more for her car than Shane paid for his car.

(a) Work out how much more Theresa paid as a percentage of the amount Shane paid.

..... %
(4)

Kylie bought a van.

After 1 year, the value of the van was \$39 865

During this year, the value of the van decreased by 15%

(b) Work out the value of the van when Kylie bought it.

\$
(3)

(Total for Question 6 is 7 marks)

YOUR WORKING

- 9 Hari scored 140 points playing a game on Tuesday.
This was 12% more points than he scored playing the game on Monday.
Work out the number of points that Hari scored playing the game on Monday.

(Total for Question 9 is 3 marks)

YOUR WORKING